Response to Referee R6

We thank the editor and referees for their continued engagement with our paper. Below we provide a brief summary of the main changes in this revision and then respond to each comment in turn.

- **Stock-return analysis.** Following R2's suggestion, the paper now presents the stock market evidence in a table and three event figures. Table 1 reports raw returns for four portfolios—the value-weighted market, equal-weighted portfolios of firms with and without gold-clause exposure, and the gold-exposure-weighted portfolio—over each of the three key events, along with firm counts. Figures 3–5 plot all four series around each event. Over the Joint Resolution window, gold-exposed firms returned 30.9–31.4% against 25.4% for comparable non-exposed firms and 12.1% for the market. A new Internet Appendix section examines every intermediate legal event of the litigation (the first hearing, the lower-court rulings, and the certiorari grants) and reports the corresponding return differentials (Table IA.20); apart from the November 1934 window, in which the second certiorari grant coincided with the midterm election and which we discuss separately, none produced significant differential returns.

- **Robustness with the full set of fixed effects.** Following R6's request, new Internet Appendix Table IA.19 re-estimates columns 2–10 of Table 7 with industry-year fixed effects, so that every characteristic-control specification is also shown with the full fixed-effect structure. The average exposure effect remains negative and significant at the 5% level in all nine specifications and $1933 \times \tilde{d}$ at the 5% level in seven of nine (10% in all nine); the $1934 \times \tilde{d}$ point estimates remain stable in magnitude but lose precision in several columns. Section 5.5 now states the underlying power constraint plainly, in addition to the mechanical collinearity that arises when the characteristic-period interactions and industry-year cells are absorbed simultaneously.

- **Limitations of the exposure measure.** Section 5.5 now contains an explicit limitations paragraph acknowledging that, because virtually all corporate bonds of the era carried gold clauses while preferred shares carried none, our exposure measure is necessarily entangled with the composition of a firm's fixed claims, and that our placebo and control strategies mitigate but cannot fully eliminate this concern.

- **Pre-trends and event-time patterns.** Section 5.2 now acknowledges directly that, given the width of the confidence intervals around individual year estimates, the litigation-year coefficients cannot be statistically distinguished from individual pre-treatment estimates such as the 1930 coefficient, and clarifies that the support for the parallel-trends assumption rests on the full pre-treatment pattern.

- **Liberty bond discussion.** We have revised the historical background section to clarify why Liberty Bond prices rose during the Supreme Court arguments in January 1935. Since Liberty Bonds were gold-denominated government obligations, a government loss would have obligated the Treasury to honor gold payments at the then-prevailing price of $35 per ounce—69% above the pre-abrogation level—increasing the real value of the bonds to holders. We have also added the missing *New York Times* citation.

First, I appreciate the improvement in measuring d_{it} relative to all long-term obligations involving fixed contractual payments, but this only partly addresses my concerns. Preferred shares are not interest bearing, and thus may behave quite differently than debt during downturns or when firms are close to distress. I understand that there is nothing the author can do about this because bank debt is simply too small, but the paper should acknowledge this limitation more clearly and briefly discuss the potential confounders that cannot be addressed due to how firms structured their debt at that time.

The referee raises a valid point, now addressed explicitly in the manuscript. Preferred dividends are discretionary in a way that bond coupons are not—deferring them does not trigger default—though the flexibility is partial, since preferred issues of the era were typically cumulative. Because preferred shares enter the denominator of $\tilde{d}$ but not the numerator, the concern is that high-$\tilde{d}$ firms, which rely more on inflexible bond financing, may be more vulnerable to downturns generally, so that the 1933–1934 differential partly reflects debt rigidity rather than gold clause litigation.

Two results argue against this. The preferred-share placebo (column 6 of Table 4) replaces the numerator of $\tilde{d}$ with preferred shares and finds positive, insignificant 1933 and 1934 coefficients, where the flexibility story predicts negative ones. And the temporal pattern (Figure 6) is concentrated in 1933–1934 and reverses after the 1935 ruling, whereas a debt-rigidity mechanism should reappear whenever conditions deteriorate—yet 1937–1938 shows no differential effect. The industry-year fixed effects and characteristic-interacted controls of Table 7 further absorb shocks correlated with industry and observable firm structure.

These tests mitigate but cannot fully eliminate the concern: in this period virtually all corporate bonds carried gold clauses and preferred shares carried none, so gold exposure cannot be separated cleanly from liability composition. Section 5 now acknowledges this limitation explicitly.

I appreciate using 1936–37 as a placebo. While this helps addressed concerns about downturns driving the results, there are other important macro differences between both periods—such as the collapse of bond issuance, that limited rollover—that may affect exposed firms more in 1933–34 beyond the uncertainty of the abrogation of the gold cause. That is, the placebo helps but only *for a subset of the concerns about confounders. To that end, the robustness regressions are more helpful. Though the findings are broadly reassuring, they do point to either the thinness of the data or a lack of robustness. The authors cannot control for industry* $\times$ *year and* $X_{it}\times$ *year at the same time, so the fixed effects vary across specifications in Table 6. The paper argues this is done "To assess robustness to firm-level controls separately from industry-level controls, this specification omits industry-year fixed effects." This sounds like a copout. Given the concerns of New Deal policies as confounders, which may have affected firms differently given their characteristics, one may want to control for both. My sense is that they lack power, as even without the industry-year FEs the significance is lost for* $1934 \times d$ *across several specifications for Columns 2–10. I would like to see the results with industry-year FEs for these columns as well. If significance is lost across the board, then the paper should acknowledge it and make its best case to defend their preferred specifications due to limited power—but readers should know whether the effects are sensitive so they can pass their own judgement.*

We thank the referee for these comments and address each in turn.

Placebo and the rollover channel. The referee correctly notes that the 1936–37 placebo addresses recession-driven confounders but not the collapse of bond issuance that may have disproportionately affected gold-exposed firms—which held more bonds—through a rollover channel (the collapse in industrial bond issuance is shown in Figure 7 of the paper). This concern is addressed directly by column 3 of Table 4, which excludes firms with bonds maturing at any point between 1931 and 1934. The 1933 and 1934 coefficients remain -0.058 and -0.037, essentially unchanged from the baseline, indicating that the investment effect is not driven by firms with immediate refinancing needs.

Industry-year fixed effects in columns 2–10. We agree that readers should be able to assess the sensitivity of the results to this choice directly. We have re-run columns 2–10 of Table 7 (Table 6 in the previous version of the manuscript) replacing year fixed effects with industry-year fixed effects, so that all ten columns share the same fixed-effect structure as column 1. The results are reported in Internet Appendix Table IA.19, whose columns (1)–(9) correspond to columns 2–10 of Table 7.

The paper's main result rests on the baseline specification of Table 4, which is already confirmed to be robust to industry-year fixed effects by column 1 of Table 7: the 1933 and 1934 coefficients are -0.059^{***} and -0.036^{***}, nearly identical to the baseline. The specifications in columns 2–10 of Table 7 are supplementary robustness checks for observable firm heterogeneity, not the preferred specification. Among those supplementary checks, adding industry-year fixed effects leaves $\tilde{d}$ negative and significant at the 5% level in all nine columns, and $1933 \times \tilde{d}$ significant at the 5% level in seven of nine columns and at the 10% level in all nine.

The referee is correct that $1934 \times \tilde{d}$ loses significance in seven of nine columns when industry-year fixed effects are added. We want to be clear, however, that this is a power problem rather than a robustness failure. The point estimates for $1934 \times \tilde{d}$ are negative and consistent in magnitude across all nine specifications, ranging from -0.012 to -0.031, with no sign reversals or attenuation. It is the standard errors that widen, not the coefficients that shrink. This pattern is characteristic of reduced identifying variation, not of a coefficient that disappears under scrutiny.

The precision loss has a mechanical explanation. In the eight portfolio decile specifications, the decile indicators are constructed from 1930 firm characteristics, and within a 2-digit industry firms tend to cluster in similar deciles; the decile-period dummies and industry-year cells therefore absorb overlapping variation in investment, widening standard errors. Because $1934 \times \tilde{d}$ has a smaller point estimate than $1933 \times \tilde{d}$, this widening is enough to push the 1934 interaction below conventional thresholds in six of these eight columns, while the 1933 interaction remains significant in all eight (at the 10% level in the cash-decile column, at the 5% level or better in the other seven). The all-controls linear specification (column 2 of Table 7, shown as column (1) of Table IA.19) is one of the two columns (along with the cash-decile specification) in which the 1933 interaction weakens the most, retaining significance only at the 10% level. Unlike the portfolio decile specs—each of which controls for one characteristic non-parametrically—this specification interacts all eight characteristics with the four-period structure simultaneously. When industry-year fixed effects are added, the after-period interactions of all eight characteristics are dropped for perfect multicollinearity. The mechanism is worth spelling out. The four period indicators (pre-1933, 1933, 1934, after) sum to one for every observation, so for any fixed 1930 characteristic the four characteristic-period interactions sum to the characteristic itself—a firm-level constant that the firm fixed effects already absorb. The interaction set therefore carries an exact linear dependency, and once the industry-year cells are absorbed as well, the after-period interaction becomes linearly determined by the remaining regressors and is dropped for each of the eight characteristics. The specification that survives estimation is no longer the fully-controlled model it was designed to be. With eight of thirty-seven controls removed and the remaining variation further divided among industry-year cells, the 1933 interaction joins the already-imprecise 1934 interaction in losing precision. In every column the point estimates retain their sign and magnitude; it is their standard errors that grow.

Column 1 of Table 7 remains our preferred specification for addressing the New Deal confounder concern. Industry-year fixed effects absorb any shock that affected an industry uniformly in a given year, which covers the canonical channels through which New Deal policies influenced investment. The characteristic controls in columns 2–10 address a distinct concern—that gold exposure may proxy for observable firm characteristics that followed differential investment trajectories—and they continue to provide reassuring evidence even when industry-year fixed effects are added, since $\tilde{d}$ remains significant throughout. We acknowledge that combining both robustness exercises simultaneously comes at a power cost, discuss this limitation explicitly in Section 5, and provide IA Table IA.19 so that readers can form their own judgment.

The authors made no comment about the fact that the 1933 and 1934 coefficients do not seem statistically different than the 1930 effects (Figure 3).

We acknowledge this observation regarding the yearly coefficient plot (Figure 3 in the previous draft; Figure 6 in the revised manuscript), and we now address it in the revised manuscript. Two points bear on it. First, the identifying contrast in our design is not between the litigation years and 1930, but between each litigation year and the omitted base year of 1932—the year immediately preceding the abrogation. Relative to 1932, the 1933 and 1934 interaction coefficients are individually significant, whereas none of the pre-treatment coefficients, including 1930, is. This is the comparison that identifies the effect; requiring 1933 to be statistically distinguishable from a separately and noisily estimated 1930 coefficient is not the contrast the design relies on. Second, the confidence intervals around individual year estimates are wide given two-way clustered standard errors in a sample of this size, so pairwise comparisons between any two individual years will often fail to reach significance.

The relevant question for parallel trends is not whether each pre-treatment year can be individually distinguished from 1933 or 1934, but whether the pre-treatment coefficients display individually significant departures from zero or a systematic trend leading into the treatment period. Neither is present. The pre-treatment coefficients in column 2 of Table 4 are +0.040, +0.026, +0.012, +0.022, −0.032, and −0.007 for 1926 through 1931, respectively, none of which is statistically significant. Crucially, these coefficients are uniformly small and individually insignificant, and the two years immediately preceding the litigation—1931 (−0.007) and the omitted 1932 (0.000)—sit essentially at zero, so there is no trend running into the treatment window that could account for the sharp 1933 effect.

The 1930 dip (−0.032, insignificant) deserves a brief comment. In 1930 the Great Depression was already deepening—bank failures were accelerating and credit conditions were tightening sharply. Since gold-exposed firms were systematically more leveraged (Table 2), they may have been somewhat more sensitive to this general credit contraction than unexposed firms, independent of gold clause exposure. The fact that this coefficient is small and insignificant, and is followed by coefficients of −0.007 and 0.000 (omitted) in 1931–1932, is more consistent with occasional noise than with a persistent pre-trend. Note also that if leverage sensitivity to downturns were the operative mechanism, it should reappear in the 1937–1938 recession—yet we find no differential effect there, which is difficult to reconcile with a leverage-based explanation of the 1933–1934 coefficients.

That said, we acknowledge that limited statistical power prevents us from rejecting the null that the 1930 coefficient equals the 1933 or 1934 coefficient. We have added a sentence to Section 5 acknowledging this directly and noting that the support for the parallel-trends assumption rests on the full pre-treatment pattern.

One small comment—my understanding (though I could be wrong) is that repurchases in the open *market were not allowed post 1929. I imagine the main reason driving the decline in shares was reorganizations. If that is the case, it is not obvious one would want to include this if the point of the analysis is to estimate whether firms were using payout policy to preemptively give the cash to shareholders. That said, the results in Table 4 show that the effects are driven by dividends, not repurchases, so this is mostly about guiding modern readers on what firm policies were in the 1930s.*

We are glad to clarify this for modern readers, and our data allow us to speak to it directly. Two facts stand out. First, treasury stock does appear on the Moody's balance sheets in our sample: between a fifth and a third of firms report a nonzero treasury-stock position during 1930–1934, and net additions to treasury stock occur in roughly one in eight firm-years. We cannot tell from the balance sheet how these shares were acquired—open-market purchases, private transactions, exchanges, or settlements—so this observation does not by itself settle the institutional question the referee raises. Whatever their source, the amounts are small: the median net addition is about 0.5% of total assets. Second, our net repurchase measure is constructed from split-adjusted changes in shares outstanding, and its variation over 1930–1934 is dominated by the issuance side: the number of shares outstanding is entirely unchanged in two-thirds of firm-years, roughly two-thirds of the measure's variation comes from episodes of net issuance, and the share reductions that do occur are modest, with a median of about 1% of market value. The referee's economic point therefore stands regardless of the institutional details: nothing in this period resembles the discretionary buyback programs of modern data, and net repurchases were not a meaningful payout vehicle.

We agree that modern readers would benefit from this context, and we have added a footnote to Section 5 stating these facts: treasury-stock additions occur but are small, the number of shares outstanding is typically unchanged, and the variation in the net repurchase measure comes mostly from share issuance rather than retirement.

This does not affect any of our substantive conclusions. As the referee notes, Table 5 (Table 4 in the previous draft) shows that the payout effect is driven entirely by cash dividends (column 2 is significant), while the net repurchase coefficient (column 3) is statistically insignificant in 1933 and 1934. The channel relevant to our debt overhang interpretation—shareholders extracting value through discretionary distributions in anticipation of losing their claim on the assets—operates through dividends, not through changes in shares outstanding.

Additional comment (cover letter)

A key issue is whether the authors can clearly disentangle the effects of uncertainty about the abrogation of gold clauses from other confounders—other policies and firm characteristics, and whether it is about the gold clause vs interest-bearing leverage. There is not much they can do about the second given how firms structured their debt. I appreciate the refinement in the definition of the treatment effect, but the ideal case would be not to have preferred shares either. I am willing to look passed this, but the paper should be more upfront. The bigger issue is other confounders. They have tried to address this in the robustness section in Table 6, but as you can see they cannot control for a full set of fixed effects. What bothers me more is that they have not shown us the results with all effects, and instead state they do not include industry-year FEs in all columns to assess the robustness to industry and firm separately. This is a weak argument. If the issue is that they have too few firms to do this without the treatment effects being killed, they should say that. Readers may understand, but they need to know. And I would like to see how bad effects get—is it noisy or do the results become small and insignificant? I also find the placebo less helpful than the paper claims, but that's a personal taste.

We are grateful for the referee's overall assessment and for the clarity of this summary. We agree with the central message—that the paper should be upfront about what the data can and cannot establish—and we have revised the manuscript accordingly. We respond to the two main issues in turn, and indicate where each is now addressed in the paper.

Gold clause exposure versus interest-bearing leverage (preferred shares). The referee is correct that the ideal design would isolate gold-clause exposure while holding the debt-versus-preferred composition fixed, and that the institutional structure of the period does not permit this: virtually all corporate bonds of the era carried gold clauses, while preferred shares carried none. We no longer leave this implicit. Section 5.5 now contains an explicit limitations paragraph stating that $\tilde{d}$

is necessarily entangled with the composition of a firm's fixed claims, that preferred-heavy firms may behave differently in distress for reasons unrelated to gold clauses, and that our tests mitigate but cannot fully eliminate this concern. We retain the preferred-share placebo (column 6 of Table 4), which finds no investment effect when the numerator of $\tilde{d}$ is replaced with preferred shares, as the most direct evidence that the effect is not driven by the general flexibility difference between bond-heavy and preferred-heavy firms.

Other confounders and the full set of fixed effects. We agree that the previous draft's justification for not always including industry-year fixed effects read as a rationalization, and we have removed that framing. We now do two things. First, we state the real reason plainly in Section 5.5: with a sample of this size, absorbing industry-year variation and characteristic-year variation simultaneously exhausts the identifying variation available to estimate the treatment effects. Second, and most importantly, we now show the results. Internet Appendix Table IA.19 re-estimates all of columns 2–10 of Table 7 with industry-year fixed effects, so the reader sees exactly "how bad effects get." The answer is that they become noisier, not small: across all nine specifications the average exposure effect $\tilde{d}$ remains negative and significant at the 5% level, $1933 \times \tilde{d}$ remains significant at the 5% level in seven of nine (and at the 10% level in all nine), and the $1934 \times \tilde{d}$ point estimates remain negative and stable in magnitude (-0.012 to -0.031, no sign reversals) even where they lose significance. This is the signature of reduced power rather than a disappearing effect, and we now say so directly rather than asking the reader to take it on faith.

On the placebo. We take the referee's point that the 1936–37 placebo addresses only a subset of confounders—chiefly recession-driven ones—and not, for example, the collapse of bond issuance (Figure 7 of the paper) and the attendant rollover risk. We have tempered the paper's language about the placebo accordingly and now lean more heavily on the direct robustness checks, including the exclusion of firms with bonds maturing during the litigation window (column 3 of Table 4), which speaks specifically to the rollover channel.

Taken together, these revisions are intended to make the paper's limitations visible to readers while preserving what we believe is the core, robust finding: gold-exposed firms reduced investment specifically during the 1933–1934 litigation window and reversed afterward, a pattern that survives industry-year fixed effects in our preferred specification and that is not reproduced by the preferred-share or bank-debt placebos.

REFERENCES